

Embryonic exposure of estrogen and BPA in zebrafish leads to ADHD-like and ASD-like phenotypes, respectively

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ARTICLE INFO

Keywords:

Estrogen
BPA
Zebrafish
Early neural development
ADHD
ASD

ABSTRACT

Exposure to the estrogenic pollutant bisphenol A (BPA) during pregnancy and early childhood is a risk factor for numerous neurodevelopmental and psychiatric disorders in humans. To understand why early BPA exposure is associated with attention deficit hyperactivity disorder (ADHD) and autism spectrum disorder (ASD), we have analyzed a series of zebrafish behaviors, neurodevelopmental process, and gene expression profiles, after a moderate level of estrogen (17 β -estradiol, E₂, as a positive control) and BPA treatments during embryogenesis (2–48 h post fertilization). E₂ exposure-caused hyperactivity was likely due to elevated expression of *cyp19a1b* since blocking aromatase activity rescued the defect. Furthermore, E₂ exposure resulted in impulsive behaviors, perhaps due to a reduced expression of brain *th* (crucial for dopamine synthesis), resembling the ADHD phenotypes. However, the hyperactivity upon BPA exposure was due to a reduction of GABAergic neurons, particularly in the midbrain. BPA-exposed fish were less-social, with increased repetitive behaviors and escape rate (during strobe light stimulation), like the ASD phenotypes. Taking advantage of published single-cell and bulk RNA-sequencing data related to zebrafish BPA exposure, we uncovered that embryonic midbrain GABAergic neurons express less *stmn1a* upon BPA exposure. When *stmn1a* function was partially lost, 14-day post-fertilization larvae became less social, further stressing the ASD phenotype after BPA exposure. Upon embryonic E₂ and BPA exposure, we have unexpectedly unveiled zebrafish ADHD-like and ASD-like phenotypes, respectively, suggesting that women of childbearing age should be cautious to use BPA and estrogen related products.

1. Introduction

Autism spectrum disorder (ASD) and attention deficit/hyperactivity disorder (ADHD) are among eight commonly seen psychiatric disorders in modern life. They are categorized as developmental defects at the neuronal level, affecting the dopaminergic, oxytocinergic, estrogenic, GABAergic, and glutamatergic signals (Crider et al., 2014; Luffe et al., 2022; Luffe et al., 2021; Mariggio et al., 2021; Sasaki et al., 2015). The causes of these disorders are largely genomic alteration-based, via de novo, acute or heritable mutation, and epigenetic changes are also being unveiled to contribute to these neurodevelopmental diseases (Shirvani-

Farsani et al., 2021; Zhong et al., 2023). ASD and ADHD are also associated with environmental pollutants, thanks to an established correlation between a drastic increase in ASD and ADHD incidents and the global impacts of rapid industrialization (Li et al., 2018; Stein et al., 2023). Since the hazardous introduction of estrogenic pollutants such as BPA and its derivatives (BPAs) more than half a century ago, the neurotoxicity of BPA is still not fully understood despite the enormous studies discussing its disruptive effects on the neurological system (Costa and Cairrao, 2024). How BPAs affect human neural development and why they lead to severe psychiatric phenotypes in patients remain elusive, partly due to a lack of evaluation of early neurogenesis of

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<https://doi.org/10.1016/j.pnpbp.2025.111293>

Received 10 April 2024; Received in revised form 10 February 2025; Accepted 18 February 2025

Available online 26 February 2025

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mammalian embryos. From the developmental biology view, the neural specification, and the birth of the first neurons start early, yet how BPAs impact such an early phase of neural development is largely not reported perhaps because of the feasibility of monitoring the developmental process in the uterus. Many studies have revealed that higher ADHD and/or ASD rates are due to high circulating level of BPAs during their mothers' pregnancy (Casas et al., 2015; Hansen et al., 2021; Harley et al., 2013). Our recent investigation/survey revealed a considerably high level of serum BPAs in white collar workers of childbearing age (Gao et al., 2021), implying a high risk of ASD and ADHD in their offspring. These correlation studies urged a thorough/systemic study on the cellular and molecular basis of the potential impact of early estrogenic exposure. To understand the cellular and/or gene profile changes underlying the BPAs-mediated development of ASD and ADHD, many non-mammalian animals, particularly the zebrafish have been used to study the pathology of the diseases (Costa and Cairrao, 2024; Hoffman et al., 2016; Lange et al., 2012; Yang et al., 2013).

The zebrafish is a relatively new model animal established in 1980–90s, due to its complete external development, large brood-size, transparent embryos, the availability of numerous genetic tools, and its high similarity to humans at the genomic and brain structure levels (Kimmel et al., 1995; Vaz et al., 2019). Moreover, from intensive studies using zebrafish embryos and larvae, behaviors caused by nervous system defects are easily captured by high-speed visualization and analysis (Shen et al., 2020). Indeed, various phenotypic/characteristic behaviors correlated to the major psychiatric disorders have been established in zebrafish, making it an attractive model for systematically evaluating the causes of most human neural diseases (Fontana et al., 2019; Meshalkina et al., 2018). As for evaluation of the impact of estrogenic chemical exposure on the brain development and function, particularly during early zebrafish embryogenesis, whether BPAs exert neurotoxic impact, regulate the neurogenesis, or initiate sex differentiation of developing brain earlier is uncertain. The gonadal and brain sex differentiations are independent of each other, with the former occurring much later (Chen and Ge, 2013). The onset of *cyp19a1b*, encoding the brain aromatase, occurs at 24hpf (hour post fertilization) during zebrafish forebrain development (Mouriec et al., 2009). Whether the onset of *cyp19a1b* marks the initial status of brain neurogenesis, the brain sexual differentiation, or both is unclear, even though zebrafish brains have not shown sex-specific differences at so early stage. Vertebrate brain estrogen can program certain brain regions in males upon aromatization of testosterone to estrogen, however, the nature of this program or re-program is unknown. Mainly expressed in radial glial cells (RGCs), which are transitioned from neuroepithelial cells after the birth of the first neurons in developing CNS, *Cyp19a1b* is associated with high neurogenic activity (Johnson et al., 2016; Mouriec et al., 2009). Thus, the presence of RGCs in the developing brain is considered as the earliest indication of brain neurogenesis, rather than a sign of male specification of brain development. As proof, Chen et al. found that BPA exposures at 8, 12, and 30hpf lead to an increased neurogenesis as a sum of altered differential processes of the neural plate border, neural crest, and neuronal cells (Chen et al., 2020). Nonetheless, easy delivery of chemicals such as BPAs to treat developing zebrafish embryos and larvae is a significant advantage over utero exposure.

Zebrafish sex is polygenically determined due to the lack of a distinct sex determination factor like the mammalian SRY/Sry (Aharon and Marlow, 2021) that may ultimately turn on *cyp19a1b* expression in the male brain. The fact that liganded (17 β -estradiol, E₂-bound) estrogen receptors and glial-specific factor(s) can work synergistically to turn on *cyp19a1b* in some forebrain regions (Diotel et al., 2010), suggests that early exposure to estrogenic chemicals would, to some extent, lead to an earlier start of embryonic neurogenesis. However, the consequence of transient/short-term early exposure to estrogenic substances is not known. After reviewing previous toxicology studies and publications, we found an inconsistent correlation between BPAs with ASD or ADHD (Costa and Cairrao, 2024). Both disorders are somewhat co-occurring

diseases, yet they could be clearly distinguished from each other by their respective diagnosis protocols in clinics (Matsuura et al., 2014; Taurines et al., 2012). Therefore, to properly establish the disease models using non-human animals, the unshared features of ASD and ADHD should be clarified in principle.

As the first step toward this goal, we take advantage of sexually undifferentiated zebrafish embryos and larvae (up to 7dpf, days post fertilization) to resolve whether early BPA or E₂ exposure similarly causes ASD-like and/or ADHD-like defects. We aimed to establish suitable behavior tests and to assign the tested behaviors with their corresponding molecular or cellular foundation. We found early BPA and E₂ exposures of two days (2–48hpf) resulted in the ASD-like and ADHD-like phenotypes, respectively. By analyzing published single-cell RNA- and bulk RNA-sequencing data (Chen et al., 2020; Saili et al., 2013), real time-PCR, in situ hybridization, examination of neuronal development and live imaging, the significant change of neuronal types, i.e., reduced GABAergic neurons and increased glutamatergic neurons in the midbrain region, is only evidenced in early BPA-exposed embryos and larvae. On the other hand, the brain aromatase and DA-dependent ADHD-like phenotypes, upon early E₂ exposure, are distinct from those of BPA-induced ASD-like behaviors. Our study assures that early BPA exposure, at a moderate concentration, results in ASD-like phenotypes largely via the non-estrogenic pathway.

2. Methods

2.1. Fish husbandry

The AB strain and transgenic zebrafish were housed at fish room Southern University of Science and Technology (Shenzhen, China). All zebrafish experiments were conducted according to the guidelines of the ARRIVE and approved by the Institutional Animal Care and Use Committee of Southern University of Science and Technology (SUSTech-1Y202402152). The adult zebrafish (AB line) were raised in 3 L tanks (10–15 fish/tank), under a 14 h:10 h light/dark cycle with 28 °C recirculating fish waters. Zebrafish embryos were collected from spawning adult zebrafish pairs and unfertilized embryos were removed under a dissecting microscope SMZ1500 (Nikon, Japan) as described (Kimmel et al., 1995; Vaz et al., 2019). All fish embryos were raised in a thermostatic incubator. For experiments using larvae, abnormal fish were removed at 5dpf. 3 L tanks (~50 larvae/tank) were used to raise 5dpf larvae until 14dpf. Since the morphological sex of zebrafish has not been determined until 28dpf, the genders of these experimental animals were indistinguishable.

2.2. Chemical exposure regime

The BPA (CAS 80-05-7) and E₂ (CAS 50-28-2) were purchased from Sigma Aldrich. The 10 g/L (43 mM) BPA and 43 mM E₂ stock solutions were prepared in 100 % DMSO respectively (CAS 67-68-5) and stored at -20 °C. Zebrafish embryos were exposed to 100 μ g/L (0.43 μ M) BPA or 0.43 μ M E₂, or 0.1 % DMSO only from 2 to 48hpf in a glass petri dish (50 embryos per dish, 10 cm petri dish (Biosharp, #BS-90-D)) for 24 h, followed by the replacement of fresh chemical solutions. At 48hpf, all treated embryos were rinsed by the embryo medium (EM) three times, fixed immediately or allowed to develop until 5dpf or 14dpf in EM.

2.3. Single cell RNA-seq analysis

The BPA single-cell RNA sequence data is available to be downloaded at the National Center for Biotechnology Information (NCBI; <https://www.ncbi.nlm.nih.gov/>) via GSE number (GSE141664). We analyzed 30hpf scRNA-seq profiles (Chen et al., 2020) using Rstudio (v.3.5.0) with Seurat (v.4.0.2) and Harmony (v.1.2.3) packages. First, we loaded the published single-cell RNA-seq data and created a Seurat object. Cell quality control was based on minimal detected genes

(default, $n = 200$). Afterwards, all data were normalized and the variable genes (default, $n = 2000$) were identified. We then reduced the dimensionality via PCA and applied Harmony to correct for the batch effects by specifying each batch information. Finally, we used FindNeighbors (Dims = 1:20) and FindClusters (Resolution = 0.9) to cluster cells being analyzed via PCA. The cell clustering results could be visualized using a UMAP plot (DimPlot) and colored (Fig. 3A). We annotated cell types according to RNA in situ expression data in ZFIN or previous studies that have already defined cell clusters through the combinations of specific gene markers (Rajan et al., 2022; Wagner et al., 2018). Note the numbers of specific neuron types were derived originally from only one data source (GSE141664) and no error bars were shown. With the FindAllMarkers (logfc.threshold = 0.5) and clusterProfiler package (v.3.16.1) using the top 50 marker genes, Go analysis would be shown.

2.4. Behavioral tests

2.4.1. Locomotion and acceleration state test

5dpf zebrafish larvae, DMSO or treatment groups, were assigned to individual wells of a 24-well plate (Corning, #3524), bathed in 500 μ L embryo medium. The 24-well plate was placed in the behavioral box (DanioVision of Noldus) for testing spontaneous movement, after 20 min acclimation, of 20 min test period in the light (100 lx). The total travel distance (cm) and acceleration were calculated by EthoVision XT13 video-tracking software.

2.4.2. Repetitive behavior test

The unusual time staying at the edge zone of a petri dish may indicate the repetitive behavior of zebrafish (Zabegalov et al., 2019). We defined the center area of each well of a 24-well plate, the diameter of 6.7 mm from the center, as the inner zone, and the rest area to be the edge zone. To test whether early chemical exposures resulted in repetitive behavior, 5dpf zebrafish were tracked automatically and the time staying in either inner or edge zone was obtained (Noldus).

2.4.3. Acoustic habituation test

Acoustic habituation is a gradually diminished response to repeated acoustic stimuli, which may indicate if neural circuits of learning and memory is normal in larva zebrafish (Zhong et al., 2023). After spontaneous locomotion and thigmotaxis tests, and a brief rest, the same 24-well plate with tested larvae were given a series of 120 tapping stimuli, set at 1 s inter-stimulus intervals (ISIs, Noldus). After the first round of tapping stimulation, the fish were allowed to rest for 30s, followed by another round of tapping stimulation. The travel distance (cm) was recorded, and the % of habituation rate was calculated (Scaramella et al., 2022).

2.4.4. Pre-pulse inhibition (PPI) test

PPI could be measured using tapping stimuli in zebrafish (Burgess and Granato, 2007; Zhong et al., 2023). 5dpf larva zebrafish were acclimated for 30 min, and a pre-pulse (weaker) acoustic tapping was applied 500 ms before a stronger tapping was given. The % PPI was obtained accordingly.

2.4.5. Freezing behavior test

The freezing behavior is a vision-based behavior test for zebrafish to react to strobe light, and the test was performed and calculated as previously described (Rennekamp et al., 2016; Shen et al., 2020). 5dpf zebrafish is more active (travel distance) during light off phase while less active (freezing behavior) in the strobe light phase normally (Fig. 4E). When the freeze index is less than 0, the fish is meant to be afraid of the strobe light. When the index is more than 0, the fish is shown to be not frozen in strobe light stimulation; and if it is 0, the fish is insensitive to the stimulation.

2.4.6. Vestibular ocular response (VOR) test

We used a self-made equipment to collect VOR data (Sun et al., 2018). The infrared video system captured changes in the projected area of eye movements during each swing (about the earth horizontal axis), spanning from positive to negative 60 degrees, and we plotted them over time. Subsequently, the values undergo Fast-Fourier-Transform (FFT) change to obtain a peak value at the stimulus frequency, which is illustrated as the VOR index or amplitude of eye movement shown in Fig. 2.

2.4.7. Light facilitation (LF) test

Light flash shortly before acoustic stimulation enhances the acoustic startle response that is mediated by dopamine release (Mu et al., 2012). Like the PPI test, the zebrafish acoustic startle response was measured either with or without light flash. The light flash was given 200 ms preceding the acoustic tapping (Noldus). The % LF was calculated according to the differences between two startle responses (net swimming distance changes) with or without light flash.

2.4.8. Dark flash response test

Dark flash stimulates zebrafish visual system (Riley et al., 2015). After 20 min light acclimation (100 lx), 5dpf zebrafish were left in dark for 1 s, and their travel distance (cm) during the light off was measured by tracking software (Noldus).

2.4.9. Social preference

We made a testing apparatus, 12 cm(L) $\times$ 8.5 cm(W) $\times$ 2 cm(H), ourselves for 14dpf larvae with two chambers. One chamber is for one tested fish and the other for 30 same-aged fish-serving as the social cue. Normal adult fish (male) usually spend most of their swimming time at the edge close to conspecific fish, and an area (5 cm $\times$ 10 cm, 5 cm equals to 1.5–2 body lengths of an average larval fish at 5dpf) was previously employed to define the conspecific sector/social zone to monitor social preference (Liu et al., 2018). We used a self-designed apparatus, shown at the bottom of Fig. 4A, to monitor how the targeted fish reacted to its conspecific fellow in 15 min after allowing them to settle down for 10 min. The social zone, 1.2 cm $\times$ 8.5 cm (white rectangles in Fig. 4H, K), was designed in the same way, i.e., 1.2 cm is about two body lengths. % time of tested fish in the social zone was measured by tracking software.

2.5. Live imaging

The transgenic lines, *Tg(gad1b:mcherry)* (Song et al., 2017) and *Tg(elavl3:EGFP)* (purchased from CZRC-CZ160) were employed in our study. 48hpf or 5dpf transgenic fish were anesthetized in 0.01 % tricaine, embedded in 1 % LMP-agarose and imaged under a Leica SP8 confocal microscope using 10 $\times$ objective. For cell counting, we zoomed in the midbrain area of each fish (re-constructed 3D image) and used Cell Counter ImageJ Plugin to count the cell number.

2.6. Whole-mount in situ hybridization (WISH)

WISH was performed as previously described (Phillips et al., 2001), and the probe sequences are listed in Table S2. After WISH embryos developed their colors, the re-fixed embryos were imaged under a microscope (Leica M205FCA). We also used ImageJ to quantify the intensities of colored embryos, pending that the entire WISH process was performed indifferently among specimens.

2.7. Loss-of-gene function study

F₀ embryos injected at one-cell stage with specific guide RNA(s) and Cas9 protein (NEB #M0646). gRNA-Cas9 F₀ somehow partially resemble gene KO individuals (Wu et al., 2018; Zhong et al., 2023). To validate whether some *stmn1a* gRNA-Cas9 F₀ could phenocopy early

BPA exposure-induced abnormal behavior(s), two gRNAs against *stmn1a* (synthesized by Genscript Co.) were used.

2.8. Quantitative RT-PCR

Total RNA was extracted from 48hpf or 5dpf zebrafish by TRIzol (Invitrogen, #15596018CN). GoScript™ Reverse Transcription kit (Promega #A5001) was used to synthesize cDNA from 0.5 µg of template RNA. The qRT-PCR was performed with an ABI-7500 using 2× qPCR Master Mix (Promega #A600A). The total RNA from each sample was normalized to β-actin and the relative transcription level was calculated with $2^{-\Delta\Delta Ct}$ method. In each experiment, three batches of fish were employed in duplicate (96-well plate?). PCR primer pairs are listed in Table S3. To test the genome-editing efficiency of gRNAs, total DNA was extracted from individual injected zebrafish and regular RT-PCR was performed using *stmn1a*-specific primer pair (Table S5).

2.9. Fluorescence-activated cell sorting (FACS)

Transgenic fish, *Tg(gad1b:mCherry)*, were exposed to DMSO, BPA or E₂ for two days (2–48hpf). Each treated group was collected at 48hpf and 5dpf. After a brief rinse with the ice-cold PBS, 40 treated embryos or larvae were passed through a needle (27G, Shengguang Medical Products Co., Ltd. #0197) for 10 times, and digested in 0.075 % Trypsin (with 0.3 mM EDTA) at 32 °C for 15 min. The enzymatic reaction was terminated by adding fetal bovine serum to a final concentration of 10 %. The resulting suspension of each group was passed through a strainer (The pore size of 40 µm, Corning, #431750) to get rid of undispersed tissues or large cell clumps. The cells with fluorescence were then analyzed by FACSCanto SORP (BD). All live cells were stained by Hoechst 33342 (YEASEN, #40731ES10) (1 µg/mL) to estimate the preparation quality and cell number. Wild-type fish without any chemical treatments were used as the negative control group, to set the gate for mCherry (red fluorescence) detection. The gate strategy is shown in Fig. S7, and the percentages of mCherry⁺ cells equal to the ratios of mCherry⁺ cell number/Hoechst⁺ cell number.

2.10. Statistics

All the data were first tested for the assumption of normality and homogeneity to determine the parametric or non-parametric data in GarphPad Prism statistical analysis. Next, the student's *t*-test or Mann-Whitney-*U* test (*U* test) for two groups comparisons and one-way analysis of variance (ANOVA) or Kruskal–Wallis (*K*–*W*) test multiply groups comparison was applied for statistic according to the normality and homogeneity of data. The data were plotted by GarphPad Prism or Rstudio. The confidence level of *p* < 0.05 was used for significance.

3. Results

3.1. E₂ but not BPA exposure results in an earlier onset of forebrain neurogenesis

We chose 0.43 µM, a mid-range concentration of E₂ and BPA (Chen et al., 2020; Moreman et al., 2017; Saili et al., 2013) to conduct our experiments and focused on the forebrain/hypothalamus region to be related to the hyperactivity. As expected, *cyp19a1b*, encoding zebrafish brain aromatase, expression responded to the administrated E₂ only (Fig. 1A, B left). When collected embryos were treated with 0.43 µM E₂ (not with control or BPA) for nearly two days (2–48hpf), a period encompassing the onset and peak of hypothalamic neurogenesis, enhanced *cyp19a1b* expression was evidenced in the forebrain by WISH and an estimated 2-fold increase of its expression was captured by real-time RT-PCR. Upon the E₂ withdrawal, strong *cyp19a1b* expression (nearly 3-fold increase) remained at 5dpf in E₂-treated fish (Fig. 1B right), unlike the control- and BPA- groups, suggesting an established auto-feedback regulation (E₂-ER-*cyp19a1b*) loop in developing brain of E₂-treated fish. We further suspected that BPA might not correlate with E₂ at all to affect the developing brain but revealed that both E₂- and BPA-treated groups exerted hyperactivity phenotype (Fig. 1C–F), which has previously been considered as the core ADHD phenotypes, in addition to inattention and/or impulsivity. No apparent difference in locomotion changes (net value) was evident between both treatment groups.

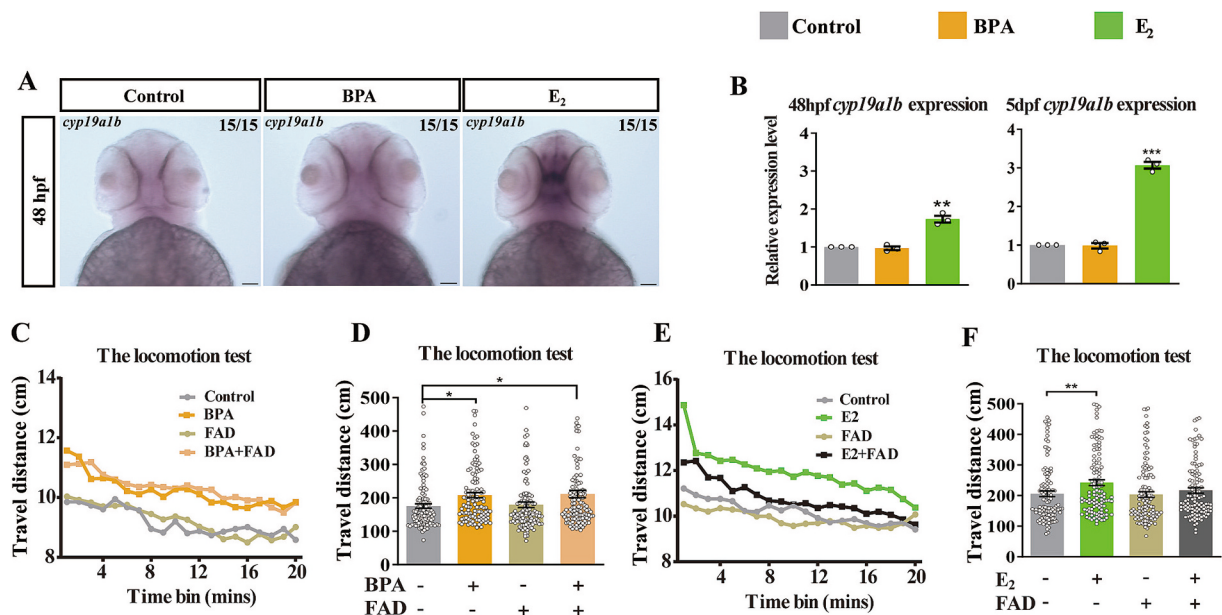


Fig. 1. Early E₂ exposure affects the forebrain neurogenesis.

(A) In situ hybridization for *cyp19a1b* expression at 48hpf, upon early E₂ or BPA exposure (*n* = 15). (B) qRT-PCR result shows the expression of *cyp19a1b* of BPA and E₂ fish at 48hpf and 5dpf (*n* = 3, one-way ANOVA with Dunnett's multiple comparisons test, 48hpf: *F* values = 52.6, 5dpf: *F* values = 331.5). (C–F) Spontaneous movement of 5dpf zebrafish larvae. DMSO, BPA-, E₂-, FAD(1 µM)-, BPA-FAD(1 µM) and E₂-FAD(1 µM) treated groups (the period of each treatment was from 2 to 48hpf). The average travel distances for each treat group are compared (*n* = 120, *K*–*W* test with Dunnett's multiple comparisons test) and differently plotted. All values were calculated as the mean value ± SEM. *: *p* < 0.05, **: *p* < 0.01.

Since BPA exposure failed to up-regulate forebrain/hypothalamus *cyp19a1b* expression, the possibility of activating *cyp19a1b* via interacting with the androgen receptor by BPA was excluded (Kinch et al., 2015). To verify if ER/E₂ mediated *cyp19a1b* expression directly led to the hyperactivity, an aromatase inhibitor, fadrozole (FAD), was used to accompany E₂ and BPA treatment. FAD eliminated E₂-induced spontaneous hyperactivity while failing to attenuate BPA-led higher locomotion (Fig. 1C–F), indicating that early E₂- and BPA-exposure induced hypothalamic aromatase-dependent and -independent hyperactivity, respectively. In other words, the hyperactivity caused by BPA exposure might be due to other brain defect(s) independent of aromatase during neurogenesis.

3.2. E₂ exposure leads to ADHD-like behaviors

To see to what extent E₂- and BPA-treated groups were different, we employed and adapted more behavior tests of zebrafish larvae and found that E₂-treated fish exerted more impulsiveness, an indication of impulsiveness (Fig. 2A). By checking the hypothalamic *th* expression at 48hpf, early E₂ treatment resulted in a significantly reduced expression (Fig. 2B). The RT-PCR results further confirmed the reduction of *th* expression (Fig. 2C). Lower tyrosine hydroxylase of hypothalamic cells is responsible for lower dopamine (DA) production in the developing brain, leading to the behavior change(s) mediated by DA. The light facilitation (LF) test, in which a visual stimulus (light flash) is applied shortly before sound/tapping-evoked startle response (Fig. 2E), is DA-dependent. As a result, the startle response is significantly enhanced, and such an enhancement is largely mediated by DA neurons in the caudal hypothalamus and D1 dopamine receptor activation/DA surge (Mu et al., 2012). A sharp reduction of %LF was evidenced in E₂-treated group (Fig. 2E), likely due to the reduced DA production (Fig. 2B). These results suggested that earlier onset of or enhanced forebrain/hypothalamus neurogenesis may affect proper dopaminergic neuron development, preventing the required DA surge triggered by light flash in the LF test.

Since ADHD patients frequently exert abnormal vestibular or sensory functions (Caldani et al., 2020), we added a test of linear vestibular-

ocular reflex (VOR) that enables larva zebrafish to stabilize gaze by rotating its eyes in the opposite direction during its body/head movement (Sun et al., 2018). At 5–8dpf, VOR is exclusively mediated by the otolithic organ (the semicircular canals are still in development and not functional) of inner ear, requiring at least three domains to sense the head position, to process the input and to execute a motor output. E₂ treatment did not cause any otic eye/vision defect, i.e., the dark flash response appeared normal (Fig. S1), and a drastically weakened VOR was validated in E₂-treated group (Fig. 2F), indicating a central processing issue. DA neurons of the hypothalamus, particularly the preoptic nucleus, are increasingly known to project too many brain areas to influence motor behavior and sensory-motor integration (Engert, 2020). Thus, the VOR circuit of E₂-treated fish is most likely interfered with by the DA deficit. Because both early E₂- and BPA-exposure resulted in hyperactivity and reduced VOR (Figs. 1C–D; 2F), but other behaviors are overall different between early E₂- and BPA-exposure groups, it's necessary to understand the underlying neural basis.

3.3. Early BPA exposure leads to abnormal neurogenesis in the midbrain

We noticed a single-cell RNA sequencing (scRNA-seq) study (Chen et al., 2020), revealing that early BPA exposure of 30 h resulted in abnormal development of the CNS, PNS, and heart. However, in our experimental setting, we observed no obvious heart and otic defects after BPA withdrawal by 48hpf, and most if not all, fish survived well into 14dpf. Nevertheless, Chen et al. revealed that genes heavily involved in the growth or regeneration of axons were identified to be down-regulated in the midbrain cells, and the pathways of nucleotide metabolism and replication and repair were significantly changed in the midbrain and differentiating neuronal cells (Chen et al., 2020). Thus, abnormal brain neurogenesis in the region(s) other than the forebrain might lead to hyperactivity (aromatase- and DA-independent) upon early BPA exposure.

Zebrafish embryos typically undergo neurogenesis around 30hpf. To understand the detailed changes in neurogenesis, particularly the neuronal fates and their shares in the total neural cell population, we re-analyzed the published scRNA-seq data (Chen et al., 2020) from NCBI

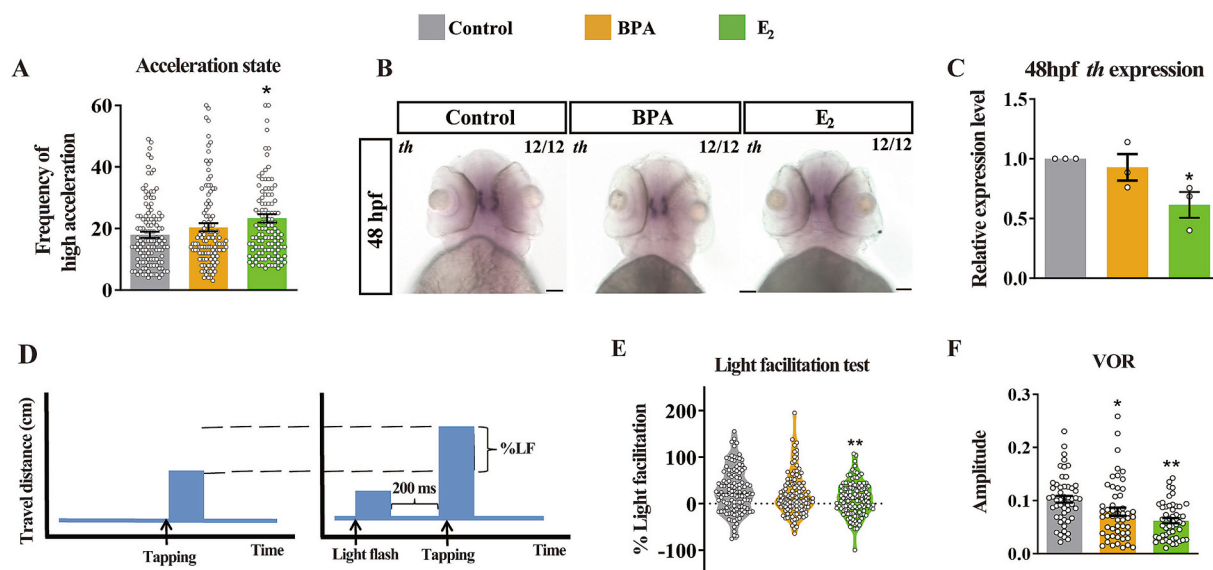


Fig. 2. E₂ exposure leads to ADHD-like phenotypes.

(A) Averaged frequencies of high acceleration state in three (DMSO, BPA- and E₂-treated) groups of 5dpf larva fish ($n = 120$, K–W test with Dunnett's multiple comparisons test). (B) The expression of *th* in 48hpf embryos of the three groups ($n = 15$ /group). (C) qRT-PCR shows that *th* expression levels of the three fish groups at 48hpf ($n = 3$, one-way ANOVA with Dunnett's multiple comparisons test, F values = 5.176). (D) Diagram of light facilitation (LF) test. The light flash applied 200 ms before acoustic tapping can enhance acoustic startle response. (E) The averaged %LF of the three groups (5dpf larva zebrafish) are shown ($n = 96$, one-way ANOVA with Dunnett's multiple comparisons test, F values = 4.832). (F) The averaged (VOR) amplitudes of the three groups ($n = 24$, K–W test with Dunnett's multiple comparisons test) are shown. All values were calculated as the mean value $\pm$ SEM. *: $p < 0.05$, **: $p < 0.01$.

(GSE141664). Using u-map and specific combinations of expressed marker genes, we identified 33 cell clusters present in both control and BPA groups at 30hpf (Rajan et al., 2022; Wagner et al., 2018). The primitive/developing brain/CNS can be roughly partitioned into the forebrain (clusters 7 and 9), midbrain (cluster 2) and hindbrain (clusters 1 and 27) (Fig. 3A; Table S1) regions. Using the *gad1b*⁺/*gad2*⁺ definition, all GABAergic neuronal specification/neuron-birth could be recognized at 30hpf. Between BPA- and control fish, one of the most dramatic changes in neurogenesis was a significant reduction of GABA neurons in the midbrain upon BPA treatment (Fig. 3B). Previous studies

already linked the reduction of GABAergic neurons to zebrafish ASD-like behaviors (Hoffman et al., 2016; Jamadagni et al., 2021), yet whether BPA-induced hyperactivity occurs due to less inhibition provided by the inhibitory neurons of midbrain remains to be established.

To verify the above ‘cell re-sorting’ result, we examined the brain fluorescence using transgenic fish line, *Tg(gad1b:mCherry)*, that labels most GABAergic neurons. Indeed, weaker red fluorescent signal and smaller amount of *gad1b*⁺ (mCherry⁺) cells were evident in the midbrain of BPA- but not E₂-treated embryos at 48hpf (Figs. 3C–D, S2A–B). In *Tg(elavl3:EGFP)* fish brain, we also observed that BPA exposure

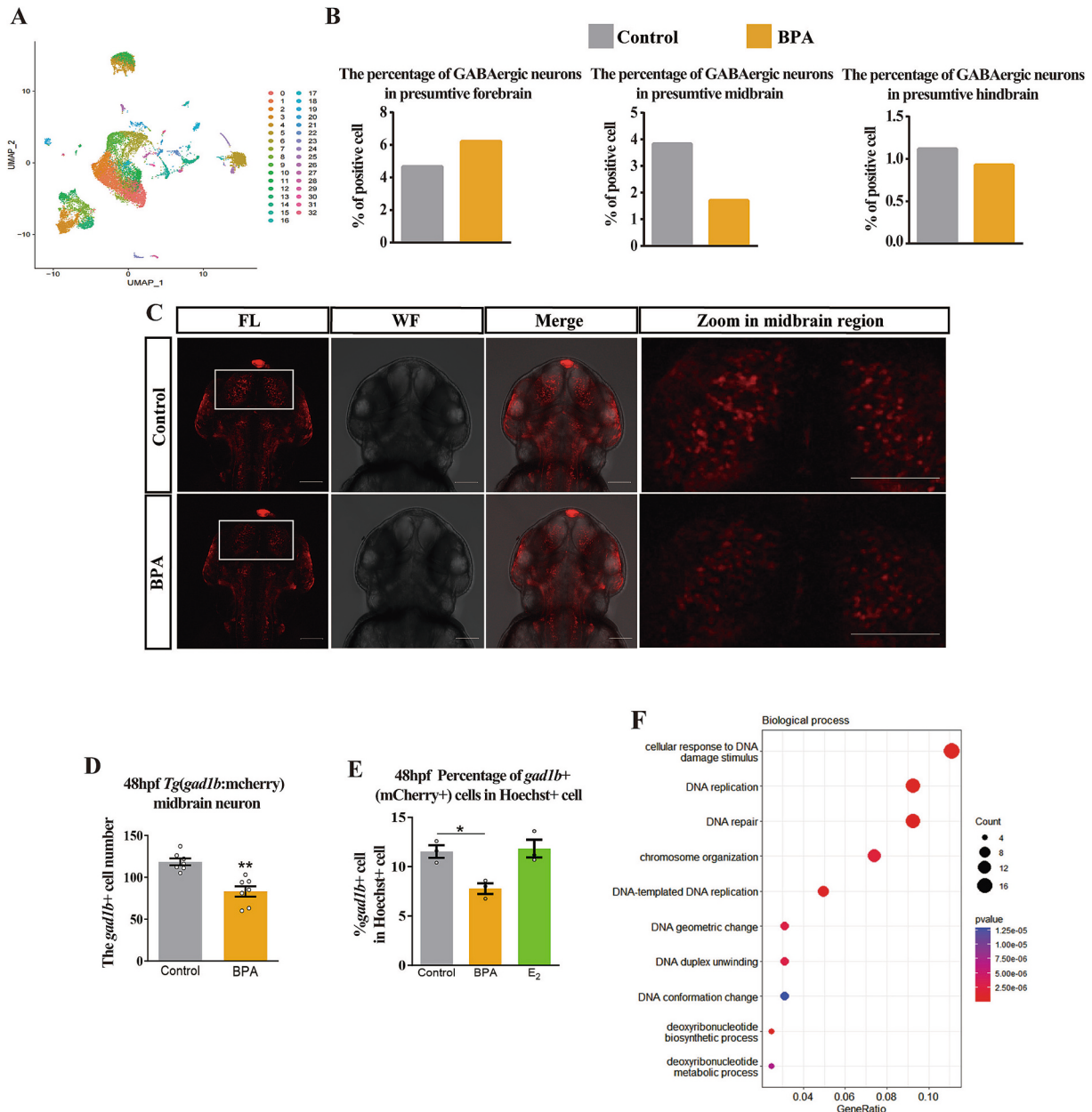


Fig. 3. Early BPA exposure affects midbrain GABAergic neurogenesis.

(A) Unsupervised clustering of the scRNA-seq data in control and BPA-exposed embryos (30hpf) reveals 33 cell/tissue clusters. (B) The change of GABAergic neurons in the forebrain, midbrain and hindbrain at 30hpf. Note the numbers of GABAergic neurons were derived originally from only one data source (GSE141664) and no error bars were shown. (C) After early BPA exposure, whole brain images show an obvious reduction of midbrain *gad1b*⁺ neurons at 48hpf. The zoom-in (outlined by the white rectangles) midbrain region was used for cell counting. All images are dorsally viewed with anterior to the top. Scale bars = 100 μ m. (D) Actual cell counting results of midbrain *gad1b*⁺ (mCherry⁺) neurons (within the white rectangles) with or without BPA exposure ($n = 7$, t-test) at 48hpf. (E) 40 embryos from DMSO, BPA-treated, or E₂-treated group (48hpf) were collected and processed for FACS analysis to quantify the percentage of *gad1b*⁺ cells. The plotted results ($n = 3$, one-way ANOVA with Dunnett’s multiple comparisons test) are shown. (F) GO analysis based on differentially expressed genes between control and BPA group in GABAergic neurons (at 30hpf). All values were calculated as the mean value $\pm$ SEM. *: $p < 0.05$, **: $p < 0.01$.

significantly affected midbrain neuronal differentiation at 48hpf while E2 exposure did not (Fig. S2C–F. *elavl3*⁺ cells are differentiated brain neurons). Therefore, a major defect caused by early BPA exposure is the abnormal GABAergic development at the midbrain.

In addition, we FACS-analyzed live cells (Hoechst staining positive cells) from 48hpf transgenic embryos exposed to DMSO, BPA or E2 and found a significant reduction of mCherry⁺ cells only in BPA-treated fish (Figs. 3E, S6A–B). In 5dpf larva fish brain, the same FACS analysis, i.e., based on mCherry⁺ cells sorted from the whole fish, still showed about 20 % reduction of *gad1b*⁺ neurons only in BPA group (Figs. S2D, S6C). These experiments indicated that early BPA exposure not only affects zebrafish neurogenesis and development but also exerts a prolonged effect on GABAergic development, which were not the case of early E2-exposure.

The neuronal types sorted out in above analysis were subjected to further analysis. For instance, the GO analysis using all midbrain GABAergic neurons revealed that the most significantly changed pathways are cellular response to DNA damage stimulus, DNA replication, DNA repair and chromosome organization (Fig. 3F), implying that BPA exposure might trigger the DNA synthesis and DNA damage-caused repair pathways. However, whether the reduction of midbrain GABAergic neurons is due to increased cell death, or an increase of thus other neuronal types in the midbrain remains to be elucidated. In addition, genes differentially expressed in the midbrain GABAergic neurons with or without BPA exposure could be picked for confirmation purposes. Nevertheless, early BPA exposure has changed the midbrain neurogenesis.

3.4. Early BPA exposure results in ASD-like behaviors

While assuring the ADHD-like phenotypes upon early E2 exposure in our hands, we utilized a series of measurements, shown in Fig. 4A, to monitor ASD- and ADHD-like behaviors side-by-side at various developmental stages. For instance, at 5dpf, the locomotion, acceleration state, repetitive behavior and freezing behavior response could be used. Also, at 5dpf, some young larvae underwent a VOR test. At 14dpf, older zebrafish larvae are subjected to social behavior tests (Fig. 4A). Same to early E2 exposure fish, early BPA exposure resulted in hyperactive zebrafish larvae, yet the degrees of hyperactivity were comparable (Fig. 4B–C). Differing from E2 group, BPA-treated fish spent more time in the edge zone in repetitive behavior test (Fig. 4D). In addition, BPA-treated fish tended to escape when they were supposed to be in a freezing mode (Fig. 4E–G). These results clearly distinguished their behaviors between early BPA and E2 exposure groups. Furthermore, BPA-treated 14dpf larvae spent less time in the social interacting zone than the time of E2 or control groups did (Fig. 4H–I), while BPA group was less social, in contrast to that of E2 fish (Fig. 2B–C). Therefore, BPA-treated fish exhibited ASD-like phenotypes, i.e., hyperactivity, abnormal repetitive behavior, impairment of social behavior.

Early BPA exposure did not significantly change % LF (Fig. 2D–E), consistent to the normal forebrain *th* expression (Fig. 2B–C). The reduced VOR (Fig. 2F) on the other hand suggested that midbrain neurogenesis defect of BPA treated group might change the function of ocular motor neurons whose cell bodies reside in the midbrain. Similarly, GABAergic neurons are involved in otolith-mediated VOR process in mouse (Shimizu et al., 2016). To further assess the extent of brain neurogenesis defect of early BPA exposure group, we employed the prepulse inhibition (PPI) and visual/auditory habituation tests, in which larva fish use neural circuits different from those of VOR and social behaviors. Both early E2 and BPA exposure groups did not exert any obvious changes (Fig. S4), indicating their normal response of sensory gating and learning memory, normal development of eye, ear and hindbrain, and normal function of related neural circuits.

The neurogenesis defect caused by early BPA exposure was rather restricted, mainly in the midbrain area, than extensive. And the defective phenotypes were distinct from those of E2 group upon a series of

examinations. We therefore speculated that early BPA exposure-led ASD-like phenotypes might be due to down-regulated expression of some genes in midbrain GABAergic neurons. *stmn1* could be one of them because it is downregulated in BPA-treated zebrafish (Saili et al., 2013). Stathmin 1, a potent microtubule-destabilizing factor to regulate microtubule dynamics, serves a role in neurite elongation and synaptic plasticity (Zhao et al., 2012). Zebrafish *stmn1a* and *stmn1b* are both expressed in the developing CNS, and *stmn1a* is indeed downregulated in 30hpf embryos upon BPA exposure (Chen et al., 2020). On the other hand, from the single-cell RNA seq database, *stmn1b* is upregulated in midbrain GABAergic neurons (Fig. 4K). When *Stmn1a* function was reduced by injecting with Cas9 and two stable guide RNAs against *stmn1a* in one-cell stage eggs (Wu et al., 2018; Zhong et al., 2023), and the F₀ larvae, sorted by the genome-editing efficiency (Fig. S5 electrophoresis gel), showed reduced social activity at 14dpf compared to those injected with Cas9 only fish (Fig. 4L). But no VOR abnormality and hyperactivity were found, suggesting that distinct changes of gene expression in midbrain GABAergic neurons are involved in and contributing to various ASD-like phenotypes. In other words, correct interneuronal connectivity and/or neural circuit requires fine regulation of microtubule dynamics that is partially mediated by Stmn1.

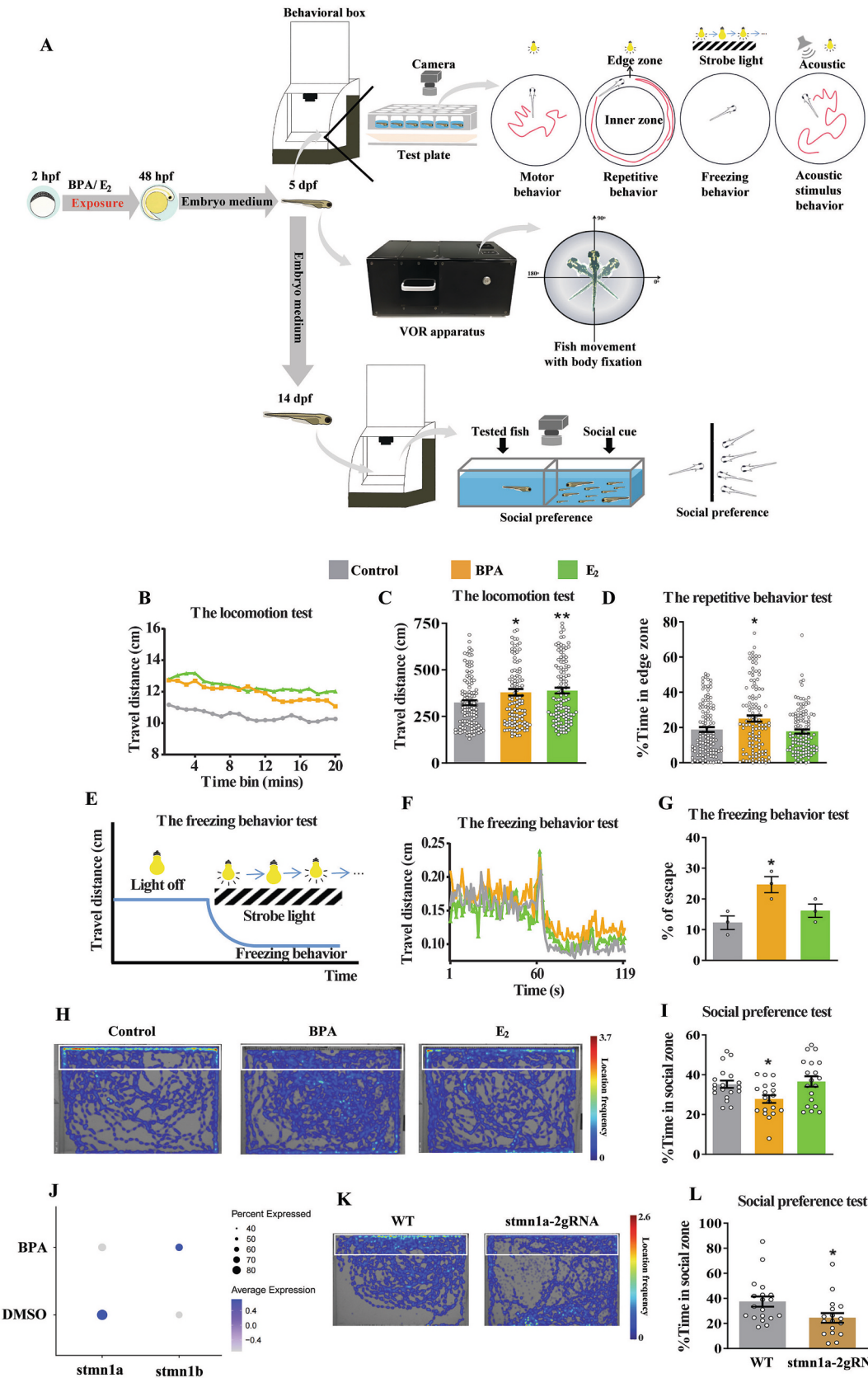
4. Discussion

Early onset ASD and ADHD are both increasing in prevalence. They commonly co-occur with each other, associated with many shared impairments including social functions. While assessing what early E2 and BPA exposure of zebrafish embryos for two days, presumably affecting the neurogenesis/neuronal differentiation rather than directly altering neural/neuronal functions, we unexpectedly found that they each brought up distinct phenotypes, which can be categorized into ADHD-like and ASD-like phenotypes, respectively, except for the ‘shared’ hyperactivity and VOR defect.

4.1. The hyperactivity induced by E2 exposure is mainly due to change in the expression level of *cyp19a1b* in the brain

We then observed that E2 exposure hyperactivity was largely due to a reduced expression of brain tyrosine hydroxylase gene, consistent with the results obtained from studying zebrafish *lphn3.1*, *per1b* and *ghrelin*. The hyperactivity of latter cases is largely due to a reduction of dopaminergic (DA) neurons when the respective gene functions are lost (Huang et al., 2015; Lange et al., 2012; Shi et al., 2021). To understand why BPA-exposed fish also exerted hyperactivity, we analyzed published RNA-sequencing data and discovered that the GABAergic neurons were significantly reduced in certain brain areas including the midbrain, responsible for the hyperactivity (Jamadagni et al., 2021; Sun et al., 2023). Detailed behavior tests and neuronal labeling further indicated that E2 group have no obvious GABAergic neuron defect, while early BPA exposure fails to enhance the forebrain neurogenesis. In other words, both treatments affect neurodevelopment differently. Both E2 and BPA exposures led to VOR impairments, however, the reduced VOR by E2 exposure was likely due to a lower level of brain dopamine, whereas the attenuation of VOR via BPA exposure was mainly due to midbrain neurogenesis defect, particularly the reduction of GABAergic neurons. BPA and *cyp19a1b* are previously correlated (Brion et al., 2012; Cano-Nicolau et al., 2016; Chung et al., 2011), but only Kinch et al. checked the consequence of BPA early exposure (8–48hpf). They found that BPA (0.0068 μM) regulates AR to adjust *cyp19a1b* function/expression, resulting in hyperactivity (Kinch et al., 2015). The E2 early exposure only was not previously checked. Therefore, upon early exposure of E2 or BPA, the brain aromatase-dependency and reduction of midbrain GABAergic neurons could help to distinguish E2-type hyperactivity and VOR defect from those of BPA-type.

BPA treatments of the embryonic, young larval and adult zebrafish have been largely studied to reveal how this small chemical affects brain



(caption on next page)

Fig. 4. Early BPA exposure results in ASD-like behaviors.

(A) The toolbox of zebrafish behavior tests for 5dpf and 14dpf larvae is employed in present study. (B–C) The averaged travel distances of three (DMSO, BPA- and E₂-treated) groups. Total travel distances of the three groups (K–W test with Dunnett's multiple comparisons test) are compared (C). (D) The averaged time staying in the edge zone for the three groups ($n = 120$, K–W test with Dunnett's multiple comparisons test) are shown. (E–G) Diagram of the freezing behavior test (E), which includes 1 min dark and 1 min strobe light binned to the 1 s intervals, and the escape (unfreezing) behavior ($n = 3$, one-way ANOVA with Dunnett's multiple comparisons test, F values = 7.3) can be identified (F–G). (H) The heat-maps show the trajectories of control, BPA- and E₂-treated fish. The while rectangles indicate the social zones. (I) The averaged time staying in the social zone (%) of the three groups (one-way ANOVA with Dunnett's multiple comparisons test, F values = 5.046) are compared. (J) In GABAergic neurons of 30hpf embryos, expression of *stmn1a* and *stmn1b* is oppositely regulated by the BPA exposure. (K–L) The heat-maps show the trajectories of WT and *stmn1a*-2gRNA fish (K), and the averaged time staying in the social zone is different (L) between the two groups of fish ($n = 17$, two-tailed U test). All values were calculated as the mean value $\pm$ SEM. *: $p < 0.05$, **: $p < 0.01$.

development and function. Regarding the duration, dosage, and observation window differences, BPA is related to numerous human diseases including ADHD and ASD. Our research has taken advantage of the zebrafish model to extend this type of research by focusing on how BPA impacts neurogenesis and defined the brain aromatase independent-defect. Our intensive analyses of published single-cell RNA-seq data pointed to abnormal midbrain neurogenesis upon early BPA exposure. Direct visualization of less GABAergic neurons in BPA fish midbrain also supported the notion that the major neurogenesis defects are different in BPA and E₂ fish brains. Our FACS analyses (Figs. 3E, S6) further indicated the prolonged impact on GABAergic development in BPA fish.

Although BPA is known to affect gene expression on a large-scale, the possibility that midbrain GABAergic neuron-expressing gene(s) may be responsible for some abnormal behavior(s) in BPA-exposed fish is reasonable. Indeed, by our analysis and gene knockdown experiments, *stmn1a* is found to be only related to social preference defect (Fig. 4K–L). The present study thus suggests that hyperactivity phenotype may be at least divided into forebrain/dopaminergic and midbrain/GABAergic types.

4.2. Exposure to BPA triggers an imbalance of E/I in the brain of juvenile fish, leading to ASD-like behavior

It is puzzling to see a high rate of comorbidity of ASD and ADHD in humans. For instance, *CHD7* and *FOXP2*, two ADHD risk/candidate genes, are often associated with ASD. Zebrafish *chd7* and *foxp2* mutants both have reduced GABAergic neurons (Jamadagni et al., 2021; Luffe et al., 2022), suggesting an intrinsic GABA connection between ASD and ADHD. In addition, mutating zebrafish homolog of a human ASD risk gene, *CNTNAP2*, results in a reduction of GABAergic neurons in the forebrain, and the *cntnap2* mutant could be rescued by estrogenic chemicals (Hoffman et al., 2016), suggesting that brain estrogen may be involved in the intrinsic GABA connection by restricting neurogenesis in the hypothalamus. Of great interest is that *cntnap2* mutant shows nighttime hyperactivity and possesses dysregulation of glutamatergic signaling, reminding us of the increased locomotion and an increase of midbrain glutamatergic neurons in BPA fish (Figs. 1E–F; 4C and S5).

From studies using peripheral biomarkers, neuroimaging, protein profiling, genetic manipulations and animal models, glutamatergic dysfunction is also evidenced. In fact, imbalanced excitatory/inhibitory (E/I) neurotransmission ratio frequently leads to ASD, and the present study showing an obvious reduction and increase of midbrain GABAergic and glutamatergic neurons, respectively (Figs. 3D–E, S2 and S5) may support such an E/I balance theory. MK-801, a non-competitive antagonist of glutamate NMDA receptors, induces the social interaction deficit in zebrafish, to be rescued by oxytocin (Zimmermann et al., 2016). In addition, estrogens act upstream of the identified NMDA and GABA deficits in *cntnap2* mutant (Hoffman et al., 2016). The brain endocrine system should be subjected to study the interaction of estrogen, GABA, glutamate and DA in behavioral circuits of ADHD and ASD. The VOR, controlled by a complex neural network that joins the vestibular pathway, multiple destinations and simultaneous outputs, defects of both treated group (Fig. 2F) also suggest the interaction and the necessity to identify and test midbrain-expressing genes that are involved in such an interaction. The reduced expression of *stmn1a* in

midbrain GABAergic neurons upon early BPA exposure (Fig. 4J) may serve such a role. We have found that partial loss of *Stmn1a* function results in abnormal social activity (Fig. 4K–L) that is likely related to ASD.

The reduced social preference/less stay in social zone in BPA group of two-week larva fish (Fig. 4I) was evident, implying the involvement of Oxytocin (Oxt) signaling that is highly associated with developmental social deficits observed in ASD animals. Oxt is a phylogenetically ancient neuropeptide that is primarily synthesized in the neurons of the hypothalamic paraventricular and supraoptic nucleus, which are near some *cyp19a1b* expression domains. By disturbing oxytocin neurons in their early life, fewer DA neurons in adult fish forebrain is responsible for their less-social phenotypes (Nunes et al., 2021), while BPA interferes with the isotocineric/oxytocineric signaling pathway to ultimately reduce oxytocin production (Naderi et al., 2022). It is therefore necessary to see if early exposure of E₂ and BPA both lead to adult social problems.

4.3. A supplementary new model distinct from the genetic mutant of neurodevelopmental diseases like ASD and ADHD

The present study provides new zebrafish models of distinct psychiatric disorders in a simpler way, reminiscent of recent trends established by well-studied animal models such as the PTZ treatment that induces acute epilepsy, MK-801 that induces schizophrenia, and VPA to model ASD in zebrafish (Baraban, 2021; Benvenuti et al., 2021; Chen et al., 2016; Gwedela et al., 2022). Although the two disorders were believed to share a similar pathogenic cause, ASD- and ADHD-like behaviors could be clearly distinguished in our experimental settings (Figs. 1, 2 and 4), so that we can treat these fish as distinct patient groups to study the underlying pathogenetic mechanisms. These new zebrafish disease models would be a significant supplement to the genetic mutants in that individual genes are disrupted. Since both ASD and ADHD are polygenetic diseases, according to numerous genome-wide association studies, abnormal pathways rather than single gene mutations are more likely to be the responsible pathogens. The combination of genetic disruptions with clear background and a wide spectrum of epigenetic modifications would provide a rich resource to understand the mechanisms underlying ASD and ADHD pathogenesis, possibly leading to clues to treat both disorders.

5. Conclusion

In the present study, we have uncovered that embryonic E₂ and BPA exposures lead to ADHD-like and ASD-like phenotypes, respectively. By a battery of behavior tests, either well-established or newly developed by us, and neurodevelopment assessments, we are in the process of understanding the neural basis of and identifying or verifying genes to be involved in the two disorders. Our current findings have established two distinct disease models in zebrafish and provided a substantial addition to the present understanding of ASD and ADHD.

Ethical Statement

The AB strain and transgenic zebrafish were housed at fish room

Southern University of Science and Technology (Shenzhen, China). All zebrafish experiments were conducted according to the guidelines of the ARRIVE and approved by the Institutional Animal Care and Use Committee of Southern University of Science and Technology (SUSTech-1Y202402152). The adult zebrafish were raised in 3 L tanks, under a 14:10 h light/dark cycle with 28 °C recirculating fish water. The zebrafish embryos were collected within 10 min between 9 and 10 am, from spawning adult zebrafish with a sex ratio of 2:2. Fertilized embryos were collected under a dissecting microscope SMZ1500 (Nikon, Japan) as described by (Kimmel et al., 1995; Vaz et al., 2019).

Funding

This work has been supported by grants from National Key Research and Development Program of China (2018YFA0801006), Natural Science Foundation of China (31771618), and Shenzhen-Hong Kong Institute of Brain Science-Shenzhen Fundamental Research Institutions (2021SHBS0002) to DL.

CRediT authorship contribution statement

Qiaosen Shen: Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Feng Zhao:** Writing – review & editing, Writing – original draft, Supervision, Investigation, Conceptualization. **Na Zhang:** Writing – review & editing, Investigation. **Ling Zheng:** Writing – review & editing, Investigation. **Dongmei Su:** Writing – review & editing, Investigation. **Yongyi Qian:** Writing – review & editing, Investigation. **Liu Xin:** Writing – review & editing, Investigation. **Sun Mingxia:** Writing – review & editing, Investigation. **Zhang Hongxu:** Writing – review & editing, Investigation. **Fangyi Chen:** Writing – review & editing, Methodology. **Wenhui Qiu:** Writing – review & editing, Methodology. **Dong Liu:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors report no biomedical financial interests or potential conflicts of interest.

Acknowledgements

We are grateful to Mr. Dandong Yuan for his comments and stimulating discussion. We thank Ms. Ning Wang of Peking University Institute of Mental Health for the insightful comments about clinical diagnosis of ADHD and ASD. We also thank Ms. Honghong Chen and SUSTech Core Research Facilities for their technical advices and help.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.pnpbp.2025.111293>.

Data availability

Data will be made available on request.

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